Assignment:

Q.6. Let $A_1, A_2, \ldots A_n$ be subsets of a non-empty set Ω. Show that the σ-field generated by $A_1, A_2, \ldots A_n$ has at most 2^{2^n} sets.

Pf:

Let A_1, A_2 be two sets.
every subset of $\sigma(A_1, A_2)$ can be written as union of $S = \{A \cap B, A \cap B^c, A^c \cap B, A^c \cap B^c\}$
So maximum number of subset can be attained by S is $2^4 = 2^{2^2}$
Now, for $A_1, A_2, \ldots A_n$, let

$S = \{ \phi, A_1 \cap A_2 \ldots A_n, A_1 \cap A_2, \ldots A_1 A_n^c, \ldots$
$\ldots \quad A_1^c \cap A_2^c \cap \ldots \cap A_n^c \}$.

desire
~~$S = \{ \phi, A_1, A_2, \ldots$~~

$S = \{ A_1 \cap A_2 \cap \ldots \cap A_n, A_1^c \cap A_2 \ldots \cap A_n,$
$A_1 \cap A_2^c \cap \ldots \cap A_n, \ldots A_1 \cap A_2 \cap \ldots \cap A_n^c$
$\vdots \quad \vdots \quad \vdots \quad \vdots$
$A_1^c \cap A_2^c \cap \ldots \cap A_n^c \}$.

No compliment
for one compliment
for all compliment

Now, all the elements of $\sigma(A_1, A_2 \ldots A_n)$ can be obtained by finite union of S.
So maximum elements $\sigma(A_1, A_2, \ldots A_n)$ can have $= 2^{|S|}$
$= 2^{2^n}$. □□.

Q.7. $f: \Omega \to \Omega'$; F σ-field on Ω'.

$f^{-1}(F) = \{f^{-1}(A) : A \in F\}$ is a σ-field in Ω.

i) for $A = \Omega' \in F$ (∵ being σ-field)

$f^{-1}(\Omega') = \Omega \in f^{-1}(F)$

ii) Let $(f^{-1}(A)) \in f^{-1}(F)$

~~Now $(f^{-1}(A))^c = \phi$~~

Now $f^{-1}(A^c) = \{\omega \in \Omega / f(\omega) \notin A\}$

$= \{\omega \in \Omega / f(\omega) \in A\}^c$

$= (f^{-1}(A))^c$

Since, F is σ-field.

if $A \in F \Rightarrow A^c \Rightarrow F$

$\Rightarrow f^{-1}(A^c) \in f^{-1}(F)$

iii) Let $f^{-1}(A_1), f^{-1}(A_2), \ldots, f^{-1}(A_n) \in f^{-1}(F)$

Now;

~~$f^{-1}(A)$~~.

Now, $f^{-1}(A_1) \cup f^{-1}(A_2) \cdots \cup f^{-1}(A_n)$

$= \{\omega \in \Omega / f(\omega) \in A_1\} \cup \{\omega \in \Omega | f(\omega) \in A_2\} \cup \ldots$
$\cdots \cup \{\omega \in \Omega | f(\omega) \in A_n\}$

$= \{\omega \in \Omega / f(\omega) \in (A_1 \cup A_2 \cup \ldots \cup A_n)\}$

$= f^{-1}(A_1 \cup A_2 \cup \ldots \cup A_n)$.

Since, $A_1 \cup A_2 \ldots \cup A_n \in F$

So

$f^{-1}(A_1 \cup A_2 \ldots \cup A_n) \in f^{-1}(F)$

So

$f^{-1}(F)$ is σ-field. □□.

Q.8 Show that open sets in $\mathbb{R}$ can be written as a disjoint union of finitely or countably many open intervals.

Solve. Let $A \subseteq \mathbb{R}$ be the open sets. define relation on A

$x, y \in A$, $x \sim y$ iff $[\min(x,y), \max(x,y)] \subset A$

Then it is equivalence relation on A.

So A do partition of A. Say

$A = \bigcup_{\alpha \in \Lambda} I_\alpha$ where I_α is equivalence

class of relation.

Now, each each I_α contains ration number, (unless I_α chose $q_\alpha \in I_\alpha$, so for each I_α, we can get q_α. define $\varphi: q_\alpha \to \mathbb{Q}$: as $\varphi(q_\alpha) = q_\alpha$ which is injective so $\{q_\alpha\}$ collection is countable.

So, I_α, $\alpha \in \Lambda$ are countble also being intervals.
So $A = \bigcup_{\alpha \in \Lambda} I_\alpha$ is union of countble intervals. □

Q.9. Show that the Borel σ-field B_R is countbly generated. i.e. there exist a countble collection C of subsets of $\mathbb{R}$, such that $\sigma(C) = B_R$.

Solve: Let $C = \{[q, \infty) : q \in \mathbb{Q}\}$
Claim
$\sigma(C) = \mathbb{R}$.
for any $x \in \mathbb{R}$,
let $q_n \in (x - \frac{1}{n}, x)$ a rational number for each $n \in \mathbb{N}$.
Then by squeeze thm: $q_n \to x$
Now: we have

$$[x, \infty) = \bigcap_{i=1}^{\infty} [q_i, \infty)$$

$\Rightarrow$ $[x, \infty) \in \sigma(C)$ $\forall$ $x \in \mathbb{R}$
Since $\sigma\{[x, \infty) : x \in \mathbb{R}\} = B_R$
So
$\sigma(C) \supseteq \sigma\{[x, \infty) : x \in \mathbb{R}\}$ — (1)
Since $C \subseteq \{[x, \infty) : x \in \mathbb{R}\}$
$\Rightarrow \sigma(C) \subseteq \sigma\{[x, \infty) : x \in \mathbb{R}\}$
$\Rightarrow \sigma(C) = B_R$; i.e. B_R generated by countble collection of subsets of $\mathbb{R}$